INCH-POUND

MIL-D-87157/3A(ER)
2 February 1989
SUPERSEDES
MIL-D-87157/3(ER)
21 December 1987

MILITARY SPECIFICATION

DISPLAYS, DIODE, LIGHT EMITTING, SOLID STATE, RED, ALPHANUMERIC, WITH ON BOARD DECODER/DRIVER

This specification is approved for use within the US Army Laboratory Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for hermetically sealed, red alpha-numeric light emitting diode displays, with CMOS IC. Two levels of product assurance are provided as specified in MIL-D-87157.

1.2 Device type. The device type shall be as follows:

Device type	Function
01	Four character, alphanumeric display in a hermetic 18-lead package, 17-segment monolithic LED; on board CMOS IC serves as ASCII data latch and decoder and LED driver. Device has dimming features.

1.2.1 Device quality level. The device quality level shall be levels A and B as defined in MIL-D-87157.

1.2.2 Luminous intensity code. Luminous intensity code shall be as follows:

Luminous intensity code	Minimum luminous intensity µcd	Maximum luminous intensity µcd
AA	275	463
A	342	575
B	425	713
C	527	884
D	654	1096
E	810	1359
X	1/	1/

Test conditions shall be as specified in group A, subgroup 7.

1.3 Physical dimensions. See figure 1.

1/ The luminous intensity code X shall not be marked on the display or its packaging. This designation is provided for use in drawings, part lists, purchase orders, and other documentation. The actual luminous intensity code shall be marked on the display and packaging.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Laboratory Command, ATTN: SLCET-R-S, Fort Monmouth, NJ 07703-5302 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

AMSC N/A FSC 5980

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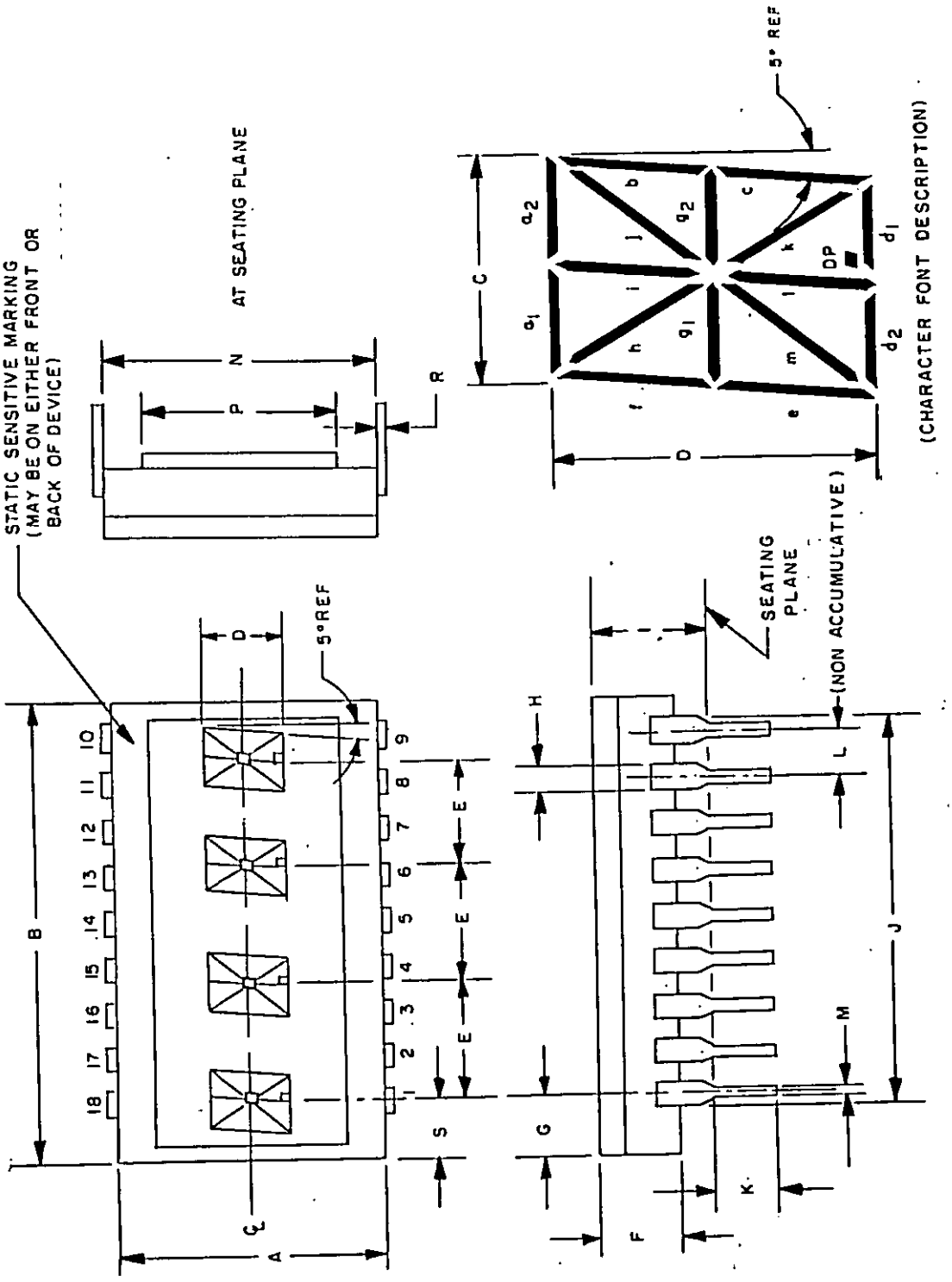


FIGURE 1. Physical dimensions.

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Symbol	Dimensions			
	Inches		Millimeters	
	Min	Max	Min	Max
A	.575	.610	14.61	15.49
B	1.080	1.115	27.43	28.32
C	.115	.145	2.92	3.68
D	.135	.165	3.43	4.19
E	.265	.285	6.73	7.24
F	.110	.215	2.79	5.46
(See note 4)				
G	.130	.165	3.30	4.19
H	.030	.055	0.76	1.40
I	.180	.260	4.57	6.60
(See note 5)				
J	.455	.870	11.55	22.10
(See note 6)				
K	.145	.190	3.68	4.83
L	.090	.110	2.29	2.79
M	.015	.025	0.38	0.64
N	.585	.615	14.86	15.62
P	.420	.485	10.67	12.32
(See note 7)				
R	.005	.015	0.13	0.38
S	.127	.147	3.23	3.73
(See note 8)				

NOTES:
1. Pin 1 shall be identified to show the starting point for lead numbering.
2. Dimensions are in inches.
3. Metric equivalents are given for information only.
4. Thickness: Top of glass to base of substrate.
5. Thickness: Top of glass to seating plane.
6. Length of glass or ceramic IC cover.
7. Width of glass or ceramic IC cover.
8. One place only.

FIGURE 1. Physical dimensions - Continued.

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1.4 Absolute maximum ratings.

Maximum total power dissipation (see 6.3):	
Device type 01 2/	918 mW
Logic supply voltage (V_{CC})	-0.5 V to 7 V dc
Voltage applied to input logic pins (see 6.3)	-0.5 V to V_{CC} +0.5 V dc
Maximum solder temperature 3/	+260°C
Ambient operating temperature 4/	-55°C to +100°C
Ambient storage temperature	-65°C to +125°C

1.5 Recommended operating conditions.

Logic supply voltage	4.5 to 5.5 V dc
Operating temperature range 4/	-55°C to +100°C
Logic input voltage, high	2 V dc minimum @ +25°C, 2.2 V dc minimum @ -55°C
Logic input voltage, low	0.8 V dc maximum

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-S-19500 - Semiconductor Devices, General Specification for.
MIL-D-87157 - Displays, Diode, Light Emitting, Solid State, General Specification for.

STANDARDS

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.
MIL-STD-883 - Test Methods and Procedures for Microelectronics.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.2 Non-Government publications. The following document(s) forms a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN NATIONAL STANDARDS INSTITUTE, INC. (ANSI)

RP-16-80 - Nomenclature and Definitions for Illuminating Engineering.

2/ Power dissipation = $V_{CC} \cdot I_{CC}$ (4 characters illuminated with 10 segments/character).
3/ $t \leq 5$ seconds at .062 inch below seating plane.
4/ $R_{\theta CA}$ = 25°C/W maximum at +100°C.

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(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated detail specifications, specification sheets, or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Detail specification. The individual item requirements shall be in accordance with MIL-D-87157, and as specified herein.

3.2 Product assurance requirements. The product assurance program and manufacturing certification requirements shall be in accordance with appendix D of MIL-S-19500.

3.3 Abbreviations, symbols and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-D-87157 and as follows:

- I_{CC} ($\overline{CU}$) - Cursor supply current.
- I_{CC} ($\overline{BL}$) - Blanking supply current.
- $\overline{BL}$ - Display blank.
- $\overline{CU}$ - Cursor select.
- I_{IH} - Logic data input current 1 state.
- I_{IL} - Logic data input current 0 state.
- CUE - Cursor enable.
- $\overline{CE}_1$ - Chip enable.
- $\overline{CE}_2$ - Chip enable.
- $\overline{WR}$ - Write.
- $\overline{CLR}$ - Clear.
- A_0 - Digit select.
- A_1 - Digit select.
- η_v - Luminous efficacy $\frac{\text{luminous power}}{\text{radiant power}}$ = 651 lumens/watt at 655 nm.
- V_I - Voltage applied to input logic.
- V_{IH} - Logic data input voltage 1 state.
- V_{IL} - Logic data input voltage 0 state.
- λ PEAK - Peak wavelength.
- λ d - Dominant wavelength.

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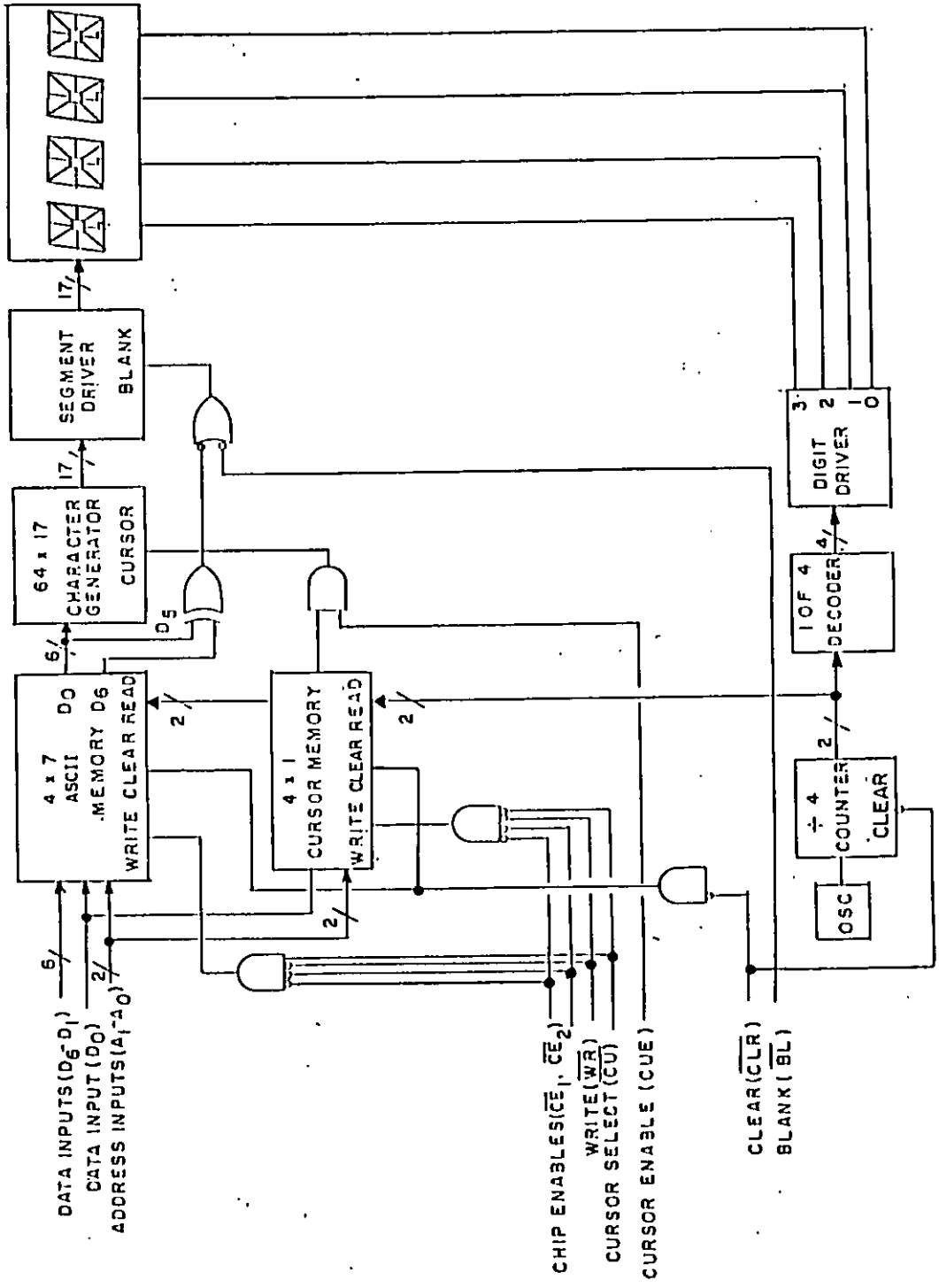


FIGURE 2. Functional block diagram, terminal connections, timing diagram, truth and character set tables.

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Terminal connections.

Pin no.	Function	Pin no.	Function
1	$\overline{CE}_1$ Chip enable	10	GND
2	$\overline{CE}_2$ Chip enable	11	D_0 Data input
3	$\overline{CLR}$ Clear	12	D_1 Data input
4	CUE Cursor enable	13	D_2 Data input
5	$\overline{CU}$ Cursor select	14	D_3 Data input
6	$\overline{WR}$ Write	15	D_6 Data input
7	A_1 Digit select	16	D_5 Data input
8	A_0 Digit select	17	D_4 Data input
9	V_{CC}	18	$\overline{DL}$ Data blank (see note)

NOTE: The above device display luminous intensity (average) can be controlled with a pulse width modulated signal on $\overline{DL}$. The imposed signal must be 2.5 kHz or higher.

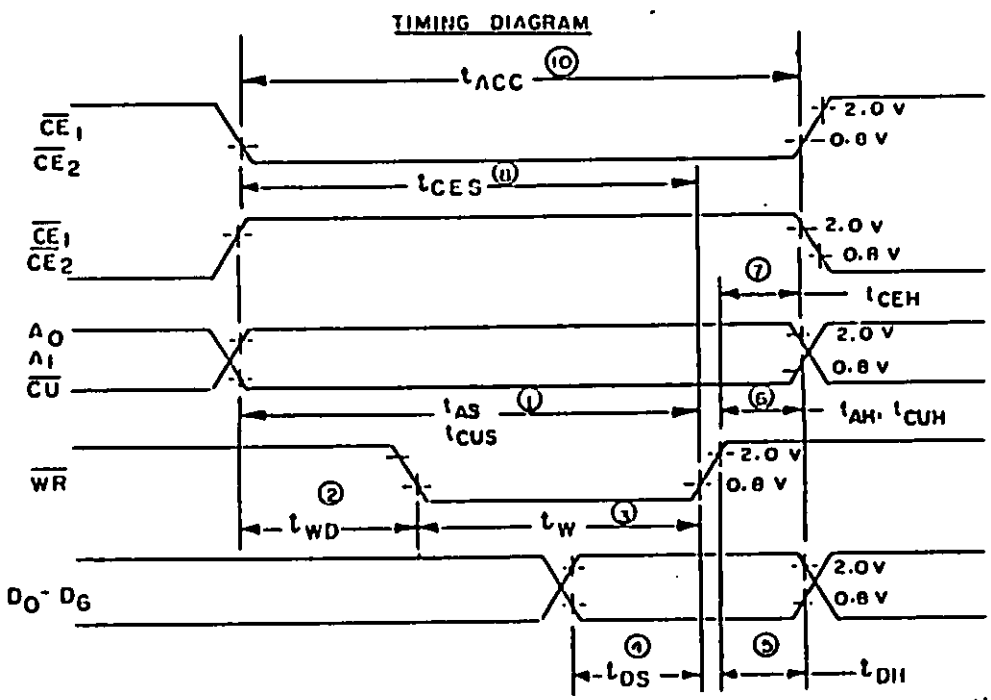


FIGURE 2. Functional block diagram, terminal connections, timing diagram, truth and character set tables - Continued.

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AC characteristics at V_{CC} = 4.5 V dc

Parameter	Symbol	-55°C(ns)	+25°C(ns)	+125°C(ns)
Chip enable setup time	t_{CES}	190	275	410
Address setup time, Cursor setup time	t_{AS} t_{CUS}	190	275	410
Chip enable hold time	t_{CEH}	25	25	25
Address hold time, Cursor hold time	t_{AH} t_{CUH}	25	25	25
Write delay time	t_{WD}	40	50	60
Write pulse	t_{W}	150	225	350
Data setup time	t_{DS}	100	150	300
Data hold time	t_{DH}	25	25	25
Clear	t_{CLR}	12 ms	15 ms	17.5 ms
Access time	t_{ACC}	215	300	435
Refresh rate (typ)		720 Hz	550 Hz	440 Hz

FIGURE 2. Functional block diagram, terminal connections, timing diagram, truth and character set tables - Continued.

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Timing diagram.

Function	$\overline{BL}$	$\overline{CLR}$	CUE	$\overline{CU}$	$\overline{CE}_1$	$\overline{CE}_2$	$\overline{WR}$	A_1	A_0	D_6	D_5	D_4	D_3	D_2	D_1	D_0	DIG_3	DIG_2	DIG_1	DIG_0
Write data memory	H	H	X	H or	L	L	L	L	L	a	a	a	a	a	a	a	NC	NC	NC	A
								L	H	b	b	b	b	b	b	b	NC	NC	B	NC
								H	L	c	c	c	c	c	c	c	NC	C	NC	NC
	X	H	X	H	L	L	L	H	H	d	d	d	d	d	d	d	D	NC	NC	NC
Disable data memory write	H	H	X	H	X	X	H	X	X	X	X	X	X	X	X	X	Previously written data			
	H	H	X	H	X	H	X													
	H	H	X	H	H	X	X													
Write cursor	H	H	H	L	L	L	L	L	L	X	X	X	X	X	X	H	NC	NC	NC	▩
								L	H	X	X	X	X	X	X	H	NC	NC	▩	NC
								H	L	X	X	X	X	X	X	H	NC	▩	NC	NC
								H	H	X	X	X	X	X	X	H	▩	NC	NC	NC
Clear cursor	H	H	H	L	L	L	L	L	L	X	X	X	X	X	X	L	NC	NC	NC	[ ]
								L	H	X	X	X	X	X	X	L	NC	NC	[ ]	NC
								H	L	X	X	X	X	X	X	L	NC	[ ]	NC	NC
								H	H	X	X	X	X	X	X	L	[ ]	NC	NC	NC
Disable data memory	H	H	H	X	X	X	H	X	X	X	X	X	X	X	X	X	Previously written data			
	H	H	H	X	X	H	X													
	H	H	H	X	H	X	X													

L = Logic low input
H = Logic high input
X = Don't care

"a" = ASCII code corresponding to symbol " A "
NC = No change
▩ = Cursor character (all segments on)

Cursor/data memory write truth table.

FIGURE 2. Functional block diagram, terminal connections, timing diagram, truth and character set tables - Continued.

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Displayed data truth table.

Function	$\overline{BL}$	$\overline{CLR}$	CUE	$\overline{CU}$	$\overline{CE}_1$	$\overline{CE}_2$	$\overline{WR}$	DIG_3	DIG_2	DIG_1	DIG_0	
Cursor enable	H H	H H	L H	X X	X X	X X	X X	A	B	C	D	Display previously written data Display previously written cursor
Clear	H	L	X	X	X	X	X*					Clear data memory and cursor memory
Blanking	L	H	X	X	X	X	X					Blank display, data, and cursor memories unchanged.

*Note: $\overline{CLR}$ should be held low for 15 ms following the last WRITE cycle to ensure all data is cleared.

ASCII character set table.

Bits	D_3 D_2 D_1 D_0	0 0 0 0	0 0 0 1	0 0 1 0	0 0 1 1	0 1 0 0	0 1 0 1	0 1 1 0	0 1 1 1	1 0 0 0	1 0 0 1	1 0 1 0	1 0 1 1	1 1 0 0	1 1 0 1	1 1 1 0	1 1 1 1
D_6 D_5 D_4	Hex	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
0 1 0	2	(Space)	!	"	#	$	%	&	'	(	)	*	+	,	-	.	/
0 1 1	3	0	1	2	3	4	5	6	7	8	9	:	;	<	=	>	?
1 0 0	4	@	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
1 0 1	5	P	Q	R	S	T	U	V	W	X	Y	Z	[	\	]	^	_

FIGURE 2. Functional block diagram, terminal connections, timing diagram, truth and character set tables - Continued.

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3.4 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-D-87157 and as shown on figure 1. Although eutectic die bonding is preferred, epoxy or organic adhesive die bonding may be performed. When epoxy or organic adhesive die bonding is used, the material, cure cycle, and other pertinent data shall be approved by the qualifying activity.

3.4.1 Terminal connections. The terminal connections and functions shall be as specified on figure 2.

3.4.2 Truth tables. The truth tables and resultant displays of the binary data in the latches shall be as specified on figure 2.

3.5 Lead material and finish.

3.5.1 Lead material. The lead material shall be in accordance with MIL-D-87157.

3.5.2 Lead finish. The lead finish shall be in accordance with MIL-D-87157.

3.6 Marking. Marking shall be in accordance with MIL-D-87157 and as specified herein. At the option of the manufacturer, the marking may be on more than one line as shown by the following example:

M87157/
00301ACX

3.7 Electrostatic discharge sensitivity (ESDS). The integrated circuit in this device is ESD sensitive. Use standard handling, storage, and assembly precautions as specified for category A devices in MIL-STD-883.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-D-87157, and as specified herein.

4.2 Screening (level A only). Screening shall be in accordance with table I of MIL-D-87157, and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see table I of MIL-D-87157)	Conditions
1.	See 4.2.1.
2.	24 hours minimum at T_A = +125°C.
3.	Condition B (10 cycles) t extreme $\geq$ 10 minutes. Transfer time $\leq$ 5 minutes.
4.	5,000 G Y1 and Y2, for 1 minute minimum.
5.	Test condition G or H maximum leak rate = 5 X 10^{-8} atm cc/s.
6.	Test condition A, C, E, or F.
7.	Subgroup 1 and 1y of subgroup 7 of table I herein.

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Screen (see table I of MIL-D-87157)	Conditions
8.	160 hours at T_A = +100°C. See 4.2.2 and figure 3.
9.	Subgroups 1 and 7 of table I herein.
10.	ΔI_V = -20 percent of pre burn-in value. ΔI_{CC} = ±10 percent.
11.	See 4.2.3

4.2.1 Internal visual (precap) inspection. Internal visual (precap) inspection shall be in accordance with method 2072 of MIL-STD-750.

4.2.2 Burn-in. The burn-in test circuits shall be as specified on figure 3. All displays shall be cooled in accordance with 4.5.1. Data input shall cycle through ASCII character set.

4.2.3 Visual and mechanical inspection. Displays shall be examined under a magnification between 3X and 10X with a field of view sufficiently large to contain the entire display. Displays that exhibit any of the following criteria shall be rejected:

a. Lead identification, markings (content, placement, and legibility), materials, construction, and workmanship not in accordance with MIL-D-87157.

b. Defects or damage resulting from manufacturing, handling, or testing.

c. Visible evidence of corrosion, contamination, or breakage (grossly bent or broken leads, cracked seals except for glass meniscus), and peeled, flaked, or blistered plating. (Discoloration of the finish shall not be cause for failure unless there is evidence of flaking, pitting, or corrosion.)

d. Leads that are not intact and aligned in their normal location, free of sharp or unspecified lead bends, and (for ribbon leads) free of twist outside the normal lead plane.

e. Leads that are not free of foreign material such as paint, or other adherent deposits, or dust.

4.3 Qualification inspection. Qualification inspection shall be in accordance with MIL-D-87157. The following additional tests shall be required for qualification approval.

4.3.1 Display color and color uniformity. The purpose of this test is to determine that the color of the display falls visually within the required color range. Two displays shall be measured in accordance with 4.5.5 to determine dominant wavelength. Measurements must fall between 625 and 660 nanometers for acceptable units.

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-D-87157 and as specified herein. If the manufacturer chooses the following option for testing, the sample displays that are to be used in group C inspection shall be designated as such prior to conducting the referenced group B test. The number of failed displays to be counted for lot acceptance or rejection as a result of the group C test shall be equal to all failed displays of the test in group B inspection which were predesignated for use in group C inspection, plus any additional failures occurring during group C testing.

a. For the steady-state operation life test in group C inspection, the manufacturer shall have the option of using all or a portion of the sample already subjected to 340 hours of group B life testing for an additional 660 hours of testing to meet the 1000-hour requirement.

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TABLE I. Group A inspection.

Inspection	MIL-STD-883		LTPD	Symbol	Limits		Unit
	Method	Conditions			Min	Max	
Subgroup 1		T_A = +25°C	5				
Logic supply current	3005	V_{CC} = 5 V dc (10 segments lighted)		I_{CC1}		115	mA dc
Logic supply current	---	V_{CC} = 5 V dc (cursor lighted)		$I_{CC1}(\overline{CU})$		165	mA dc
Blanking current	---	V_{CC} = 5 V dc (blank)		$I_{CC1}(\overline{BL})$		4.0	mA dc
Logic current "0" state	3009	V_{CC} = 5 V dc; V_{IL} = 0.8 V dc		I_{IL1}		160	µA dc
Subgroup 2		T_A = +100°C	7				
Logic supply current	3005	V_{CC} = 5 V dc (10 segments lighted)		I_{CC2}		90	mA dc
Logic supply current	---	V_{CC} = 5 V dc (cursor lighted)		$I_{CC2}(\overline{CU})$		130	mA dc
Blanking current	---	V_{CC} = 5 V dc (blank)		$I_{CC2}(\overline{BL})$		3.5	mA dc
Logic current "0" state	3009	V_{CC} = 5 V dc; V_{IL} = 0.8 V dc		I_{IL2}		100	µA dc
Subgroup 3		T_A = -55°C	7				
Logic supply current	3005	V_{CC} = 5 V dc (10 segments lighted)		I_{CC3}		167	mA dc
Logic supply current	---	V_{CC} = 5 V dc (cursor lighted)		$I_{CC3}(\overline{CU})$		225	mA dc
Blanking current	---	V_{CC} = 5 V dc (blank)		$I_{CC3}(\overline{BL})$		8	mA dc
Logic current "0" state	3009	V_{CC} = 5 V dc; V_{IL} = 0.8 V dc		I_{IL3}		180	µA dc
Subgroups 4, 5, and 6							
Not applicable							

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TABLE I. Group A inspection - Continued.

Inspection	MIL-STD-883		LTPD	Symbol	Limits		Unit
	Method	Conditions			Min	Max	
Subgroup 7			5				
Time average luminous intensity per digit average	---	V_{CC} = 5 V dc, 8 segments on, 4 digits on. Readings = (4 digits on/4)		Iy	275	1359	µcd
Functional test in accordance with truth table of figure 2	---	Logic low level = 0.8 V dc Logic high level = 2 V dc					
Subgroup 8			7				
External visual	---						

b. Group C inspection shall be performed when the first lot is produced and thereafter, every 6 months during production.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-D-87157 and table I herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table III of MIL-D-87157 and table II herein.

4.4.3 Group C inspecton. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table IV of MIL-D-87157 and table III herein.

4.4.4 Inspection of packaging. Inspection of packaging shall be in accordance with MIL-D-87157.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

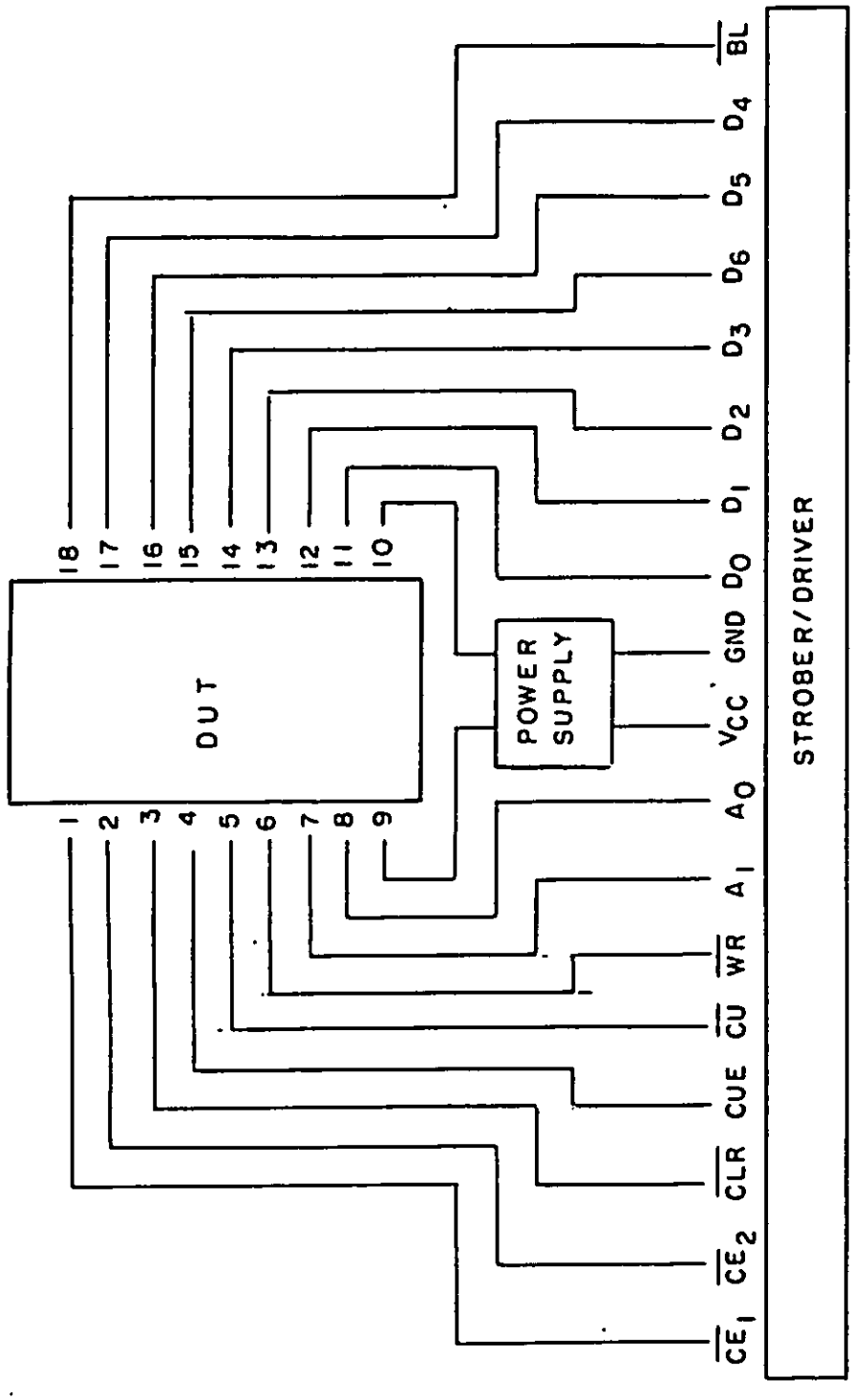
4.5.1 Life test and burn-in cool down procedure. The displays shall be cooled to room temperature prior to the removal of bias. The interruption of bias for up to 1 minute for the purpose of moving the displays to cool down positions separate from the chamber within which life testing or burn-in was performed shall not be considered removal of bias. Alternately, the bias may be removed during cooling provided the case temperature of displays under test is reduced to a maximum of +35°C within 30 minutes after the removal of the test conditions.

4.5.2 Lead integrity. Each lead of each test display shall be bent inward through an angle sufficient to cause the lead to retain a permanent bend (after stress removal) of at least 15 degrees measured at the lead extremities. At the completion of the initial bend, the leads shall be returned to their approximate original position. This procedure shall be repeated for a total of three complete bends.

4.5.2.1 Failure criteria. After removal of the stress, any evidence of breakage, loosening, or relative motion between the lead (terminal) and the display body shall be considered a failure. Meniscus cracks shall not be cause for rejection.

4.5.3 Moisture resistance. Moisture resistance shall be conducted in accordance with method 1021 of MIL-STD-750 and 4.5.3.1, 4.5.3.2, and 4.5.3.3 herein. Initial conditioning shall be in accordance with 4.5.2 herein, except only one bend shall be required.

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T_A = +100°C, V_{CC} = 5.5 V dc, time = 160 hours, all four digits to be continuously sequenced through its ASCII character set.

FIGURE 3. Burn-in circuit.

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TABLE II. Group B inspection (classes A and B of MIL-D-87157).

Inspection	MIL-STD-750		Sampling plan
	Method	Conditions	
Subgroup 1			
Resistance to solvents	1022		4 devices/ 0 failures
Internal visual and mechanical	2075	Inspection may be performed through glass cover, includes front and back cavities	1 device/ 0 failures
Subgroup 2 1/ 2/			
Solderability	2026	Immersion within .062 inch of seating plane; solder bath temperature = 245 ±5°C	LTPD = 15
Subgroup 3			
Thermal shock (temperature cycling)	1051	Test condition B1, 15 minutes dwell	LTPD = 15
Moisture resistance 3/ Visual inspection end points	1021	Within 24 hours after completion of moisture resistance test (see 4.5.3)	
Hermetic seal	1071		
Fine leak		Test condition H	
Gross leak		Test condition C	
Electrical/optical end points 4/	---	I_{CC} (10 segments on), $I_{CC}(\overline{CU})$, $I_{CC}(\overline{BL})$, I_{IL}, I_V at V_{CC} = 5 V and visual function T_A = +25°C	
Subgroup 4			
Operating life test (340 hours)	1027	T_A = +100°C at V_{CC} = 5.5 V	LTPD = 10
Electrical/optical end points 4/	---	Same as subgroup 3	
Subgroup 5			
Nonoperating storage life test (340 hours)	1032	T_A = +125°C	LTPD = 10
Electrical/optical end points	---	Same as subgroup 3	

1/ Whenever electrical/optical tests are not required as end points, electrical rejects may be used.
2/ The LTPD applies to the number of leads inspected except in no case shall less than three displays be used to provide the number of leads required.
3/ Initial conditioning shall be a 15 degree inward bend and back to original position, one cycle.
4/ Limits and conditions are per the electrical/optical characteristics.

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TABLE III. Group C inspection (classes A and B of MIL-D-87157).

Inspection	MIL-STD-750		Sampling plan
	Method	Conditions	
Subgroup 1			
Physical dimensions	2066		2 devices/ 0 failures
Subgroup 2			
Lead integrity 1/ 2/	2004	Test condition B2	LTPD = 15
Hermetic seal	1071		
Fine leak		Test condition H	
Gross leak		Test condition C	
Subgroup 3			
Shock	2016	1500 g's; Time = 0.5 ms, 5 blows in each orientation X_1, Y_1, Z_1.	LTPD = 15
Vibration, variable frequency	2056		
Constant acceleration	2006	5,000 g's at Y, orientation	
External visual 3/	or 1010 1011		
Electrical/optical end points 4/	---	I_{CC} (10 segments on), $I_{CC}(\overline{CU})$, $I_{CC}(\overline{BL})$, I_{IL}, I_V at V_{CC} = 5 V and visual function	
		T_A = +25°C	
Subgroup 4 5/ 6/			
Salt atmosphere	1041		LTPD = 15
External visual 3/	or 1010 1011		
Subgroup 5			
Bond strength 7/	2037	Test condition A	LTPD = 20 (c = 0)

See footnotes at end of table.

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TABLE III. Group C inspection (classes A and B of MIL-D-87157).

Inspection	MIL-STD-750		Sampling plan
	Method	Conditions	
Subgroup 6			
Operating life test 8/	1026	T_A = +100°C at V_{CC} = 5.5 V dc	λ = 10
Electrical/optical end points 4/	---	Same as subgroup 3	

1/ The LTPD applies to the number of leads inspected except in no case shall less than three displays be used to provide the number of leads required.
2/ MIL-STD-883 test method applies.
3/ Visual requirements shall be as specified in MIL-STD-883, method 1010 or 1011.
4/ Limits and conditions are per the electrical/optical characteristics.
5/ Whenever electrical/optical tests are not required as end points, electrical rejects may be used.
6/ Solderability samples shall not be used.
7/ Displays may be selected prior to seal.
8/ If a given inspection lot undergoing group B inspection has been selected to satisfy group C inspection requirements, the 340-hour life tests may be continued on test to 1000 hours in order to satisfy the group C life test requirements. In such cases, either the 340-hour end-point measurement shall be made a basis for group B lot acceptance or the 1000-hour end-point measurement shall be used as the basis for both group B and group C acceptance.

4.5.3.1 Failure criteria. Any display with illegible markings or which, when using a magnification of 10X to 20X, shows evidence of corrosion, flaking, or pitting of the finish that will interfere with the application of the device, shall be considered a failure. The finish shall include the entire exposed lead area from the meniscus to the tip and all other exposed metal surfaces. Markings shall include all specified markings.

4.5.3.2 Finish. When viewed at 10X to 20X magnification, any lead breakage, deep penetration of metal elements (penetration either over an area or at a single point which is greater than 50 percent of the diameter, width, or thickness), peeling or blistering which indicates loss of adhesion of the finish to the base metal or any significant formation of corrosion products (especially any bridging between unconnected metal elements by products of corrosion) shall constitute a failure.

4.5.3.3 Marking. Any specified markings which are missing in whole or in part, faded, smeared, blurred, or shifted (dislodged) to the extent that they cannot be readily identified from a distance of at least 6 inches with normal room lighting and without the aid of magnification, or with a viewer having a magnification no greater than 3X, shall constitute a failure.

4.5.4 Axial luminous intensity.

4.5.4.1 Apparatus. The equipment for this measurement shall consist of a photometer designed to respond to incident luminous flux density. The output of the photometer shall be linearly related to luminous incidence over the range of levels encountered in calibration and measurement. The output may be a voltage or a current or may be rendered directly in the units of luminous incidence.

4.5.4.1.1 Spectral response. The relative response of the photometer shall be within ±6 percent of the photopic luminous efficacy as defined by the American National Standards Institute RP-16-80 for all wavelengths within the spectrum of the device to be measured.

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4.5.4.1.2 Receptance pattern. The off-axis receptance of the photometer shall be constant over a large enough angle so that it responds equally to light from all parts of the display to be measured. An effective plane of receptance (image of the detecting surface) shall be defined with respect to which the calibration can be performed.

4.5.4.2 Calibration of photometer (see figure 4). The photometer shall be calibrated with a National Bureau of Standards (NBS) Detector Response Transfer Intercomparison Package (DRIP), an OSRAM FCR 6333 100 W, 12 V, 3300 K quartz tungsten halogen incandescent lamp or equivalent, and Omega Optical full blocking 10 nanometer bandwidth interference filters or equivalent. Peak transmission wavelengths shall be centered within the 550 nm to 740 nm range. The correlation factors derived by comparing the response of the photometer with the calculated photometric response of the NBS DRIP unit shall be used as the photometer calibration.

4.5.4.2.1 Calibration procedure. The equipment shall be mounted on an optical bench and a photo detector reference plane established. A baffle box shall be used to collimate the light and support the interference filters. The exit aperture shall be placed as close to the photo detectors as possible and shall form a spot of light of known area (A_0) that underfills the active areas of the photometer and DRIP unit photo detectors. The distance between the quartz lamp filament and the reference plane shall be within 31.4 inches (800 nm) and 35.4 inches (900 nm).

4.5.4.2.2 Equipment calibration measurements and calculation. The quartz lamp shall be illuminated and 1 hour shall be allowed for stabilization. The photocurrent output in amperes of the NBS DRIP unit and the luminous incidence by the photometer in lumens per square meter shall be measured at the reference plane for each interference filter used. The responsivity at the peak wavelength shall be determined for each interference filter used. The radiometric incidence of the NBS DRIP unit at the peak wavelength of each interference filter shall be calculated as follows:

$$E_{e_o}(\lambda_p) = \frac{I_{p_o}(\lambda_p)}{A_o R_o(\lambda_p)}$$

where $E_{e_o}(\lambda_p)$ = Radiometric incidence in watts/m^2 at peak wavelength of interference filter.

I_{p_o} = Photocurrent in amperes at peak wavelength of interference filter.

A_o = Illuminated active area of the detectors used during calibration, (m^2).

R_o = Responsivity in amperes per watt of the NBS DRIP unit at the peak wavelength of the specified interference filter.

Luminous efficacy in lumens per watt for each interference filter used shall be calculated as follows using one nanometer intervals across the filter transmission spectrum:

$$\eta_v = 683 \frac{\int T_{FILTER}(\lambda)\,\bar{y}(\lambda)\,d\lambda}{\int T_{FILTER}(\lambda)\,d\lambda}$$

where η_v = luminous efficacy in lumens/watt.

$T(\lambda)$ = relative spectral transmission of an interference filter.

$\bar{y}(\lambda)$ = 1931 luminosity function (CIE photopic curve).

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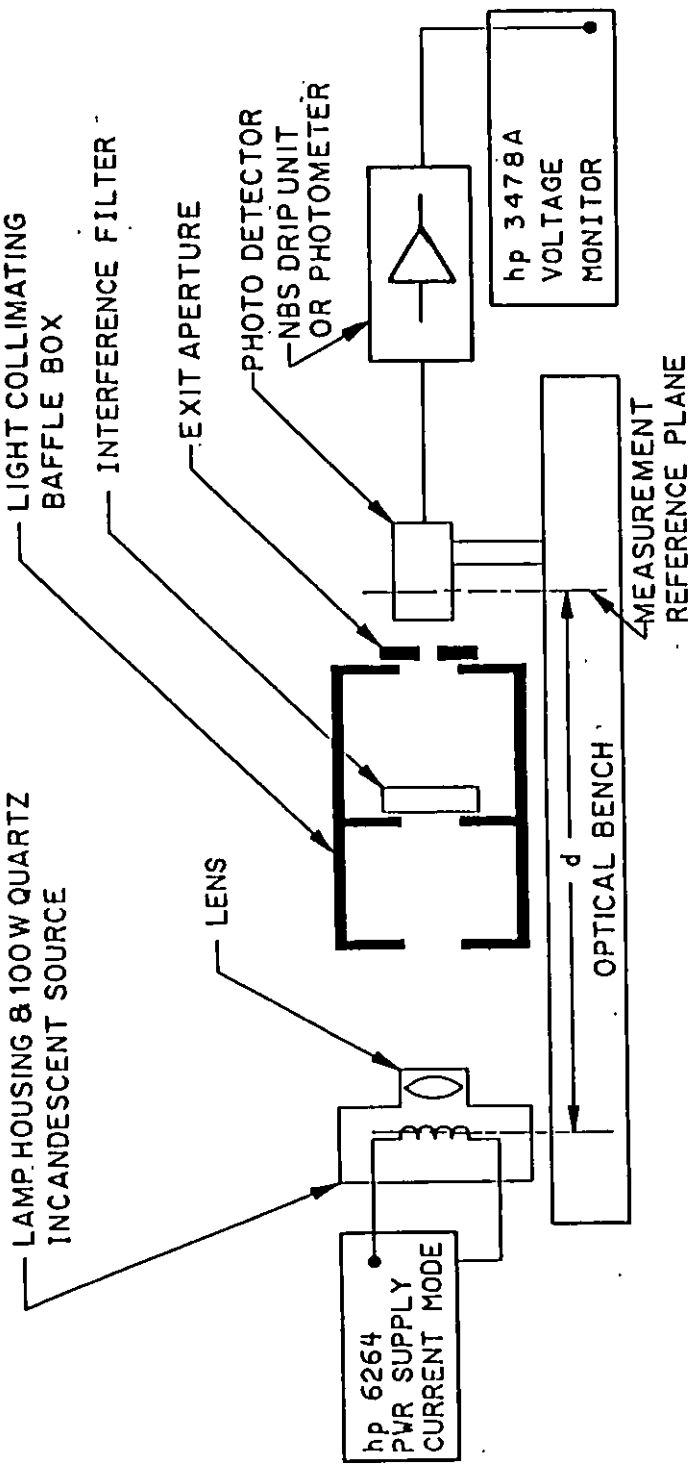


FIGURE 4. Photometer calibration block diagram.

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The photometer correlation factor at the peak wavelength for each interference filter shall be calculated as follows:

$$K(\lambda_p) = \frac{E_v\ (\lambda_p\ \text{- measured})}{E_{e_o}(\lambda_p) \cdot \eta_v}$$

where $K_{(\lambda_p)}$ = photometer correlation factor at the peak wavelength for each interference filter.

$E_v\ (\lambda_p$ - measured) = measured luminous incidence by the photometer in lumens/m^2.

$E_{e_o}(\lambda_p) \cdot \eta_v$ = calculated photometric response of the NBS DRIP unit.

The values of $K_{(\lambda p)}$ vs λ_p shall be plotted graphically and the correlation value for 655 nanometers (K_{655}) shall be determined.

4.5.4.3 Operation of photometer. The photometer shall be adjusted to measure on axis light output as luminous intensity in candelas. The correct value for luminous intensity shall be calculated by using the following equation:

$$I_v = \frac{1}{K_{655}} \quad \text{Photometer reading.}$$

where I_v = corrected value of photometer output in candelas.

K_{655} = photometer correlation factor at 655 nanometers.

4.5.5 Display color, dominant wavelength.

4.5.5.1 Apparatus. A spectroradiometer, designed to measure the spectral distribution of light sources, shall be used to measure the spectal distribution of the display under test. The spectroradiometer shall have a spectral bandwidth of 5 nanometers or less, a sensitivity of 2.4×10^{-9} watts/steradin per nanometer at 655 nanometers and a flat relative response within ±5 percent from 550 to 730 nanometers.

4.5.5.2 Calibration. Calibration of the spectroradiometer shall be accomplished by using a light source of known spectral output and dividing the reading of the spectroradiometer at each wavelength by the output of the light source at that wavelength. By generating a table of weighting factors at each wavelength of the spectroradiometer, a uniform reading at all wavelengths for a light source of uniform output may be obtained by multiplying the reading of the spectroradiometer at each wavelength by the appropriate weighting factor.

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4.5.5.3 Procedure. The LED display to be measured shall be mounted perpendicular to the entrance aperture of the spectroradiometer at a distance such that the flux emitted by all of the light emitting diodes is detected. The display shall be spectrally scanned between the wavelengths of 550 and 740 nanometers and the resulting spectrum normalized to 1.00 at the peak wavelength. The 1931 CIE x, y color coordinates may be derived by first integrating the normalized LED display spectrum to obtain the two tristimulus values X, Y and then calculating the 1931 chromaticity coordinates as follows:

$$X = \int_{550}^{740} f(\lambda)\, \bar{x}_\lambda\, d_\lambda$$

$$Y = \int_{550}^{740} f(\lambda)\, \bar{y}_\lambda\, d_\lambda$$

where $\bar{x}_\lambda$ and $\bar{y}_\lambda$ are the 1931 CIE color matching functions

and f_λ is the normalized radiated spectrum.

The integration shall be performed at a maximum of 5 nanometer intervals to determine color coordinates. The 1931 chromaticity coordinates for light emitting diodes may be calculated directly from the values of X and Y as determined above, using the following relationships:

$$x = \frac{X}{X + Y} \; ; \quad y = \frac{Y}{X + Y}$$

4.5.5.4 Derivation of dominant wavelength. The dominant wavelength may be derived within an accuracy of ±1/2 nanometer by plotting the x, y chromaticity coordinates on the 1931 CIE chromaticity diagram and by drawing a line from the CIE illuminant C point (6500°K color temperature point) through the x, y color point and intersecting the perimeter of the chromaticity diagram. The point where the line intersects the perimeter is the dominant wavelength.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-D-87157.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Notes. The notes specified in MIL-D-87157 are applicable to this specification.

6.2 Ordering data. The contract or purchase order may specify the inspection data (see MIL-D-87157).

6.3 Applied voltage limitation. A negative voltage greater than -0.5 V dc shall not be applied to any input or V_{CC} as the excessive negative voltage may damage the on-board integrated circuit.

6.4 Color binning. Color binning of the red LED displays shall not be required because the human eye is not able to discern small color differences in the red region where the radiated spectrum of these light-emitting diodes is located.

6.5 Part or Identifying Number (PIN). The complete part number shall be in accordance with MIL-D-87157.

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6.6 Subject term (key word) listing.

Alphanumeric displays
Light emitting diode displays
Optoelectronic displays, LED

6.7 Substitutability. Displays covered by this specification are substitutable for the following commercial device type:

Device type	Commercial type
01	MDL 2416C

6.8 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

CONCLUDING MATERIAL

Preparing activity:
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Agent:
DLA - ES

(Project 5980-A007)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-D-87157/3A(ER)	DISPLAYS, DIODE, LIGHT EMITTING, SOLID STATE, RED ALPHANUMERIC, WITH ON BOARD DECODER/DRIVER

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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